

Module 1: Linear Algebra (Intermediate & Advanced Topics)

1. Rank and Nullity

- Rank Determination: Rank of a matrix A is the number of linearly independent rows or columns.
- Row-Echelon Method: Use Gaussian elimination to reduce matrix to row-echelon form. Number of non-zero rows equals rank.
- LDL Decomposition: In LDL decomposition, rank equals number of non-zero diagonal elements of D .
- Rank-Nullity Theorem: For an $m \times n$ matrix, $\text{Rank}(A) + \text{Nullity}(A) = n$.
- Free Variables: Number of free variables = Total columns – Rank.

2. Determinants

- 2x2 Matrix: For $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$, $\det(A) = ad - bc$.
- 3x3 Matrix: Determinant calculated using minors and cofactors.
- Linear Dependence: If determinant = 0, vectors are linearly dependent.
- Invertibility: A square matrix is invertible iff determinant $\neq 0$ (full rank).
- Product Property: Determinant of diagonal matrix = product of diagonal entries.

3. Basis Verification

- Linear Independence: Equation $c_1v_1 + c_2v_2 + \dots + c_nv_n = 0$ must have only trivial solution.
- Spanning: Every vector in space must be expressible as linear combination of given vectors.
- Standard Basis of $\mathbb{R}^n$: Unit vectors $e_1, e_2, \dots, e_n$.
- Shortcut: If n vectors in n -dimensional space have determinant $\neq 0$, they form a basis.

4. Linear Transformations (Matrix Form)

- Matrix Representation: Every linear transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ can be represented by an $m \times n$ matrix.
- Columns: Correspond to dimension of input space.

- Rows: Correspond to dimension of output space.
- Example: $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ is represented by a 3×2 matrix.

5. Identity and Special Matrices

- Identity Matrix (I): Rank equals its dimension n .
- Diagonal Matrix: Powers computed by raising diagonal elements individually.
- Inverse of Diagonal Matrix: Replace each diagonal element by its reciprocal (if non-zero).
- Diagonalizable Matrix: A matrix A is diagonalizable if $A = PDP^{-1}$ where eigenvectors form a basis.